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Isometric back muscle endurance: An EMG study on the criterion validity of the Ito test

Roland Müller, Kurt Strässle, Brigitte Wirth *

Institute of Human Movement Sciences and Sport, ETH Zurich, Wolfgang-Pauli-Str. 27, 8093 Zurich, Switzerland

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ABSTRACT

The validity of the Sorensen test as a measure for back muscle endurance is controversial due to a possible impact of hip extensor muscles. The aim of this study was to investigate the criterion validity of an alternative test (Ito test) compared to the Sorensen test. Both procedures were performed by 29 healthy subjects (11 women) for 5 s and until exhaustion (randomized order). EMG activity was measured from 3 lumbar back and 3 hip extensor muscles. Muscular involvement in test positions was calculated as percentage of maximal voluntary contraction (MVC). Muscle fatigue was determined by the normalized regression coefficient of the median frequencies of the EMG power spectrum (NMF_{slope}). Prediction of holding time by NMF_{slope} values was investigated using regression analysis. In the test positions, the hamstring muscles were activated to a higher MVC percentage in the Sorensen than in the Ito test, while the iliocostalis muscle was less activated. Similarly, the iliocostalis ($p = 0.006$) and the multifidi muscles ($p = 0.03$) significantly contributed to predict holding time in the Ito test, whereas the multifidi muscles ($p = 0.001$) and the semitendinosus muscle ($p = 0.046$) did so in the Sorensen test. The results of this study indicate that the Ito test might present a valuable alternative for testing back muscle endurance in LBP patients.

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1. Introduction

A deficiency in isometric lower back muscle endurance is supposed to be a major risk factor for non-specific low back pain (LBP) (Jorgensen and Nicolaisen, 1987; Holmstrom et al., 1992; Demoulin et al., 2006). Beyond the variety of tests for the assessment of isometric lower back muscle endurance, the Sorensen test enjoys the most support from the available literature and is thus currently regarded as the gold standard for measuring isometric lower back muscle endurance (Moreau et al., 2001; Demoulin et al., 2006). It is easy to perform, requires no special equipment and showed high intertester and intratester reliability (Simmonds et al., 1998; Latimer et al., 1999). In terms of validity, however, a significant contribution of the hip extensor muscles to the holding time in the Sorensen test was reported (Moffroid et al., 1994; Kankaanpaa et al., 1998). Furthermore, an early cessation of the test due to spinal pain or pain in legs or abdomen is a common disadvantage of the Sorensen test (Biering-Sorensen, 1984; Latimer et al., 1999).

In 1996, Ito described an alternative test for evaluating isometric lower back muscle endurance (Ito et al., 1996). In the initial position, the subject lies in prone position with a pad under the abdomen and arms at the sides. For the test, the subject lifts the sternum off the examination table while flexing the neck as much as possible and contracting the gluteus maximus muscle for stabilizing the pelvis (Ito et al., 1996). Maximal neck flexion and gluteal muscle contraction are supposed to be the most optimal position for decreasing lumbar lordosis and for activating the lumbar erector spinae muscle to the maximum (Shirado et al., 1995). The Ito test is thus hypothesized to result in less spinal loading than the Sorensen test (Demoulin et al., 2006). In comparison to the Sorensen test, the Ito test is also easy to perform and does not require any equipment apart from a pad. Its test–retest reliability was reported to be very high without inducing any pain (Ito et al., 1996). Furthermore, its discriminative validity was proven in 190 healthy subjects and subjects with chronic LBP, respectively (Ito et al., 1996). Lastly, since the ankles are not fixated in the Ito test in contrast to the Sorensen test, the activation of the hip extensor muscles is supposed to be smaller than in the latter (Demoulin et al., 2006). However, no data on the criterion validity of the Ito test are available (Moreau et al., 2001). Apart from deficient standardization of the test procedure (pad, amount of upper body lifting), the absence of any proof of criterion validity might be one reason for the limited international recognition of the Ito test and the request for further investigations (Demoulin et al., 2006).

For all these reasons, the aim of the present study was to determine the criterion validity of the Ito test in healthy adults by

* Corresponding author. Tel.: +41 44 632 59 81; fax: +41 44 632 11 42.
E-mail address: brigitte.wirth@move.biol.ethz.ch (B. Wirth).

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comparing the Ito and the Sorensen test (1) in terms of lower back muscle and hip extensor activity in each test position and (2) in terms of fatigue of the aforementioned muscles during performance of the Ito and the Sorensen test until exhaustion. Thereby, the testing procedure of the Ito test was standardized with regard to the applied pad and the amount of upper body lifting.

2. Methods

2.1. Subjects

Twenty-nine participants (mean age 25.1 ± 1.9 years, mean BMI 22.5 ± 1.9 kg/m^2, 11 women) with no previous history of back problems were tested. All procedures were in accordance with the Declaration of Helsinki and were approved by the local ethics committee. All subjects gave written informed consent for participation.

2.2. Experimental procedure

After appropriate skin preparation with an abrasive gel (SPES Medica, Battipaglia, Italy) for the purpose of reducing skin impedance, the oval surface electrodes with a circular electrical surface contact of 1 cm^2 (Ag/AgCl, wet gel, Ambu Blue Sensor N, Ballerup, Denmark) were placed with an interelectrode distance of 20 mm according to the European Recommendations for Surface Electromyography (Hermens et al., 1999) on the right longissimus (2 finger width lateral from the processus spinosus of L1), iliocostalis (1 finger width medial from the line from the posterior spinal iliaca superior to the lowest point of the lower rib, at the level of L2), multifidi (2–3 cm from the midline at the level of L5, placed on and aligned with a line from caudal tip posterior spina iliaca superior to the interspace between L1 and L2 interspace), gluteus maximus (at 50% on the line between the sacral vertebrae and the greater trochanter), biceps femoris (at 50% on the line between the ischial tuberosity and the lateral epicondyle of the tibia) and semitendinosus muscles (at 50% on the line between the ischial tuberosity and the medial epicondyle of the tibia). After a warm-up of 5 min duration on a rowing machine (Concept2, Morrisville, USA), maximal voluntary contraction (MVC) was determined at the beginning of the test session for the back muscles, the gluteal muscle and the leg muscles in order to be able to normalize the EMG data. For the back muscles, the subjects lay in a prone position with their hands folded in the neck and maximally contracted the corresponding muscles against the resistance of the examiner at the shoulder blades (Freiwald et al., 2007). For the gluteal muscle, the subjects performed an isometric maximal contraction in a prone position with the leg extended without external resistance (Freiwald et al., 2007). For the biceps and semitendinosus muscle, lastly, the subjects lay prone with the knee in about 60 degrees flexion. The subjects were instructed to maximally contract the muscles against the resistance of the examiner (Freiwald et al., 2007). Per muscle, the subjects had one test to become familiar with the testing procedure before performing three trials. In each trial, the maximal isometric contraction was hold for 5 s (Schlumberger and Schmidtbleicher, 2000; Freiwald et al., 2007). During all tests, no verbal motivation was given.

For the purpose of comparing muscle activity in the basic position of the Ito and the Sorensen test, the participants then performed each test position for 5 s in a randomized order. After a short break of 5 min, the subjects performed, in the same random order, either the Ito or the Sorensen test until exhaustion or until the maximal time according to the corresponding test protocol (Biering-Sorensen, 1984; Ito et al., 1996). Apart from randomization, a break of 1 h between the two tests should have ensured that muscle fatigue did not interfere with test performance (Latimer et al., 1999). The performance of both tests at the same time was essential in order to guarantee identical conditions in terms of electrode placement and conductance.

2.3. Assessments

The Sorensen test was performed according to the original study by Biering-Sorensen (1984). The subjects lay prone on an examination table with the iliac crest on the edge of the table. The lower body was fixed to the table by three straps around the pelvis, the knees and the ankles (Fig. 1A). Before the beginning of the test, the participants were allowed to rest the upper body on a chair in front of the examination table. They were then asked to lift the upper trunk from the chair, to place the arms across the chest and to maintain the trunk in neutral alignment for as long as possible. Neutral alignment was monitored by means of an inclinometer, which was placed between the scapulae, at the level of the vertebra T7 (Moffroid et al., 1994) and had been calibrated previously when the subject laid prone. When the participant's body alignment deviated more than ±10 degrees from the horizontal line during the test, the examiner requested the participant to regain neutral alignment. If the neutral alignment could no longer be adopted, the test was terminated by the examiner (Latimer et al., 1999; Verna et al., 2002). Similarly, the procedure was stopped early if the participant was in pain. If neither deviation from neutral alignment nor pain occurred, the test was stopped by the examiner after maximal test time of 240 s (Biering-Sorensen, 1984).

The Ito test was conducted analogous to the original study by Ito (Ito et al., 1996). The participants lay prone on a therapy table

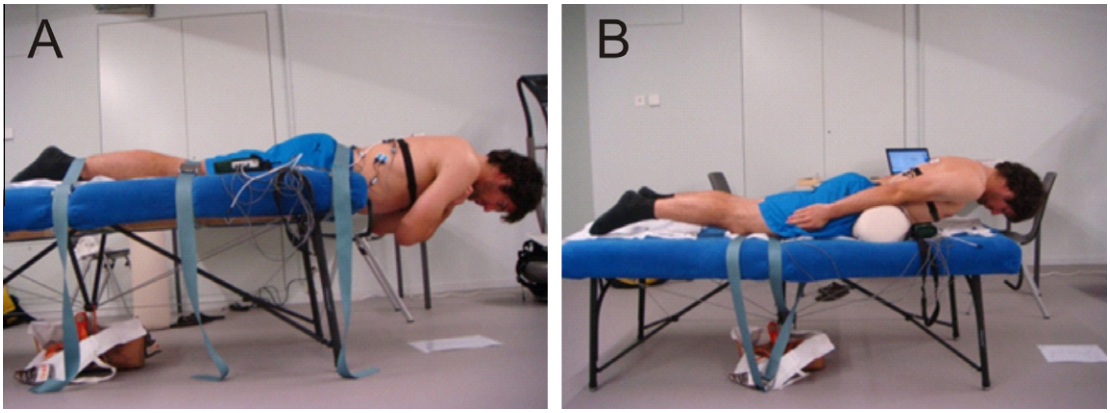

Fig. 1. Test methods. The test position of the Sorensen test (A) and the Ito test (B).

without any fixation to the table. A foam pad of 0.2 m diameter (Nudischer medcompany, Messstetten, Germany) was placed under the lower abdomen with the iliac crest on its center. The participants were requested to lift the sternum off the examination table with the cervical spine maximally flexed and the gluteal muscle contracted to ensure pelvic stabilization. The arms were held parallel to the body axis without any contact to the pad (Fig. 1B). The constraining factor of the sternum being off the examination table resulted in an angle between the upper body and the horizontal line of about 15 degrees; this was controlled by the examiner analogous to the Sorensen test, which means that a deviation of more than ±10 degrees was announced by the examiner. Maximal holding time in the Ito test was 300 s (Ito et al., 1996).

2.4. Data analysis

For the EMG measurements, a telemetric EMG was used (Telemyo 2400 T, Noraxon Inc, Scottsdale, USA). The sampling rate was 1500 Hz and the raw EMG signals were band-pass filtered between 10 and 500 Hz (input impedance: >100 MOhm, common mode rejection ratio: >100 dB, gain preamplifier: 500).

All EMG analyses were conducted using MyoResearch XP Master Package Version 1.07.17 (Noraxon Inc., Scottsdale, USA). In the data of the MVC testing and the measurement of EMG activity in the basic test positions, interference with the heart beat was reduced first from all channels by the ECG reduction tool (combination of pattern recognition and adaptive filtering (Noraxon Inc., Scottsdale, USA)). The signals were then smoothed by means of a moving RMS window (time window 500 ms). In the MVC data, maximal amplitude of each of the three bursts was determined and the average was calculated in order to account for variability. From the five seconds in the basic test position, a period of 4 s, which started when the subject was in a stable position, i.e. the inclinometer data did not show any fluctuation, was analyzed by calculating the mean amplitude over the time period. This mean activity was then expressed as a percentage of the corresponding MVC value. For the determination of fatigue, a shift of the EMG power spectrum to lower frequencies during a fatiguing muscle contraction and correspondingly a high correlation of median frequency (MF) to endurance time has been well documented (Merletti et al., 1992; Mannion and Dolan, 1994; Mannion et al., 1997). Thus, in the data of test performance until exhaustion, EMG power spectral density and its MF was computed for 1 s intervals based on a fast fourier transformation and the slope of the MF regression line over test time was determined (Mannion et al., 1997). Because of possible differences in subcutaneous layers between subjects and between different muscles of a subject, the slope of the MF regression line was, according to Coorevits et al. (2008), normalized to the initial MF, represented by the intercept of the MF regression line ($NMF_{slope} = (MF_{slope}/MF_{init}) \times 100\%$). Due to poor quality of the EMG signal in 3 MVC measurements (2 ♀, 1 ♂), muscle activity in the basic test positions could not be determined for three participants ($n = 26$), while all data could be included for the calculation of the NMF_{slope} values during test performance until exhaustion ($n = 29$).

2.5. Statistical analyses

A Mann–Whitney U Test was used to analyze the possible influence of the order of test execution and gender on holding time and EMG parameters. The correlation between the holding time in the Sorensen test and the Ito test was investigated using Spearman correlation analysis. For comparing muscle activity (expressed as a percentage of MVC) between the two basic test positions, non-parametric Wilcoxon signed-rank tests were used. The NMF_{slope} values were compared between the Ito and the Sorensen test for each muscle, also using Wilcoxon signed-rank tests. For determining the contribution of each muscle to the test holding time, a linear multiple regression analysis (enter method, all variables introduced into regression equation) was conducted. For all analyses, SPSS Statistics 17.0 (SPSS, Chicago, Illinois) was used and the significance level α was set at 0.05.

3. Results

The mean holding time was 202.3 s (±45.0 s) in the Sorensen test and 249.8 s (±64.1 s) in the Ito test, respectively. In the Soren-

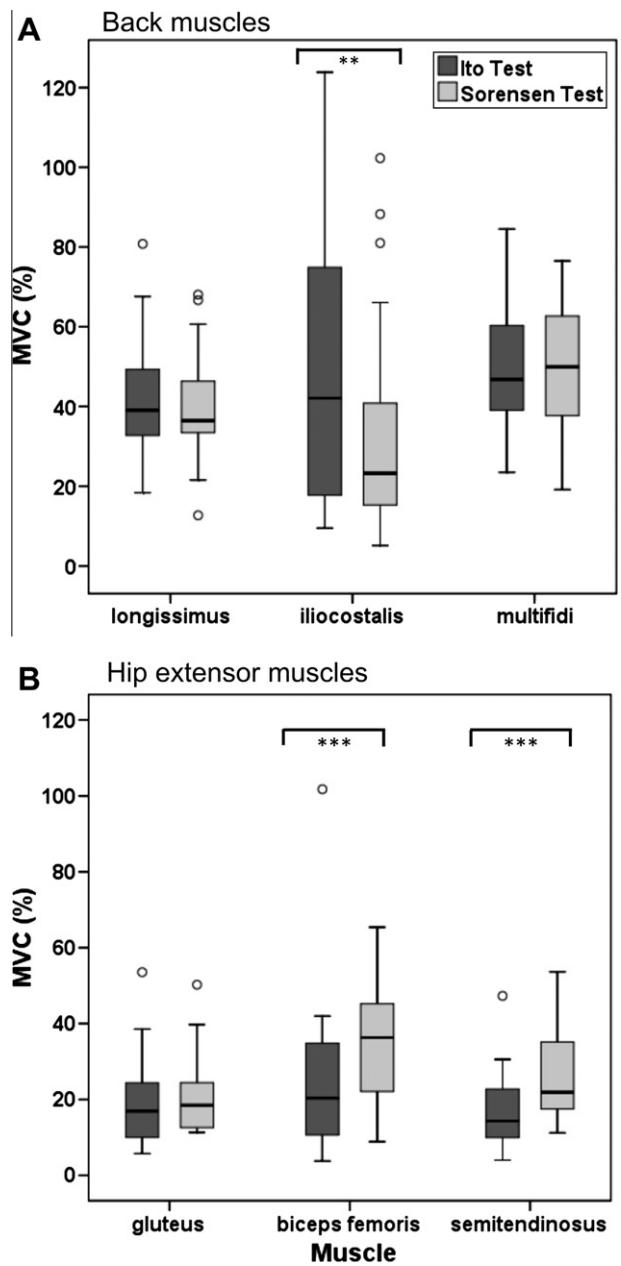

Fig. 2. Muscle activities in the two test positions ($n = 26$). The muscle activities of the back (A) and hip extensor muscles (B) in the two test positions expressed as percentages of the maximal voluntary contraction (MVC). $^{**}p < 0.01$, $^{***}p < 0.001$. ○, outliers (values between 1.5 and 3 interquartile ranges from the end of the box).

sen test, 12 subjects achieved the maximal holding time of 240 s and 17 subjects stopped early because of muscular exhaustion. In the Ito test, 11 subjects could hold the position for the maximal time of 300 s and 17 subjects terminated the procedure ahead of time due to exhaustion, one subject stopping because of lumbar pain. The correlation between the holding time in the Sorensen test and the Ito test was $r_s = 0.52$ ($p = 0.004$). The order of test execution (Sorensen test or Ito test first) did not significantly affect holding time ($p = 0.11$ for the Sorensen test and $p = 0.58$ for the Ito test). Gender, however, significantly influenced maximum holding time with longer holding times for women in both tests ($p = 0.01$ for the Sorensen and the Ito test).

3.1. Muscle activity in test position

The comparison of muscle activity in the two test positions revealed a significantly higher activity of the biceps femoris muscle ($p < 0.001$) and the semitendinosus muscle ($p < 0.001$) in the Sorensen test position, while the iliocostalis muscle showed higher activity in the Ito test position ($p = 0.005$) (Fig. 2A and B). The activity of the other muscles did not differ between the two test positions (longissimus: $p = 0.50$, multifidi: $p = 0.64$, gluteus maximus: $p = 0.71$).

3.2. Muscle fatigue during test execution

Muscle fatigue was significantly higher during the Sorensen test than during the Ito test in all muscles ($p \leqslant 0.001$ for longissimus, multifidi, biceps femoris, semitendinosus; $p = 0.002$ for iliocostalis and $p = 0.005$ for gluteus maximus) (Fig. 3) and was not affected by the order of test execution. Women had throughout higher (less negative) NMF_{slope} values indicating less fatigue, which was, however, significant only for the multifidi muscles in both, the Sorensen ($p = 0.01$) and the Ito test ($p = 0.005$) and for the gluteal muscle in the Ito test ($p = 0.03$).

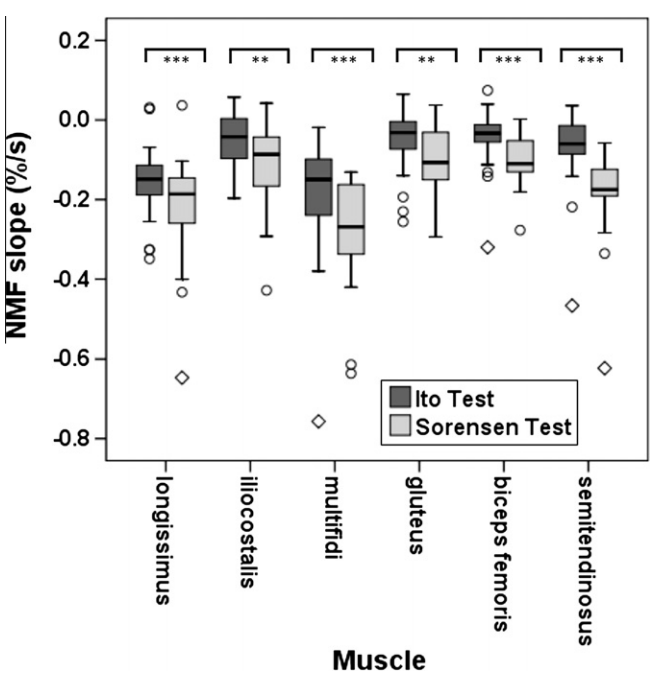

Fig. 3. Muscle fatigue during test performance until exhaustion ($n = 29$). The slope of the regression line of the median frequencies normalized to the initial median frequency (NMF_{slope}) for all muscles during test performance until exhaustion. $^{**}p < 0.01$, $^{***}p < 0.001$. ○, outliers (values between 1.5 and 3 interquartile ranges from the end of the box); ◇, extreme values (values more than 3 interquartile ranges from the end of a box).

Table 1
Influence of muscle fatigue on holding time.

Model	Regression coefficient	Significancy
Sorensen test ($R^2 = 0.73$, $p < 0.001$)		
Constant	281.1	0.000
NMF_{slope} multifidi	264.6	0.001
NMF_{slope} semitendinosus	161.5	0.046
NMF_{slope} biceps femoris	−183.2	0.171
NMF_{slope} iliocostalis	−82.3	0.329
NMF_{slope} longissimus	26.4	0.697
NMF_{slope} gluteus	−11.8	0.894
Ito test ($R^2 = 0.65$, $p < 0.001$)		
Constant	318.0	0.000
NMF_{slope} iliocostalis	486.9	0.006
NMF_{slope} multifidi	249.0	0.028
NMF_{slope} gluteus	170.6	0.286
NMF_{slope} longissimus	−102.9	0.473
NMF_{slope} biceps femoris	100.9	0.566
NMF_{slope} semitendinosus	0.4	0.998

3.3. Influence of muscle fatigue on holding time

Linear regression analysis revealed that in the Sorensen test, fatigue (NMF_{slope}) of the multifidi muscles and the semitendinosus muscle significantly influenced the holding time (Table 1), whereas in the Ito test, fatigue of the iliocostalis muscle and the multifidi muscles significantly predicted holding time (Table 1).

4. Discussion

The results of the present study showed that in the test position of the Sorensen test, more activity of some hip extensor muscles (hamstrings) was required than in the position of the Ito test. This was also reflected by the test performance until exhaustion: in the Ito test, prediction of holding time was restricted to back muscles, whereas in the Sorensen test, one hip extensor muscle also contributed to predicting endurance time. In general, the Sorensen test emerged as being more exhausting than the Ito test.

The differences in muscle activity in the two test positions might be ascribed to the stabilization of the legs by three straps in the Sorensen test (Plamondon et al., 2002; Demoulin et al., 2006). The finding of the hip extensors contributing to endurance time in the Sorensen test is in accordance with previous studies (Moffroid et al., 1994; Kankaanpaa et al., 1998). However, these results cannot fully be compared, since in the latter studies, correlation analyses were conducted, while in the present study, regression analysis was performed. Two studies that also used regression analysis found controversial results which somewhat conflicted with those of the present study (Mannion and Dolan, 1994; Coorevits et al., 2008). Mannion and Dolan (1994) reported that the most fatigable muscle was the best predictor of endurance time in the Sorensen test, while Coorevits et al. (2008) found that the best prediction of endurance time was made by the muscles that were least fatigued. The latter study, however, reported in 20 subjects that solely the thoracic part of the iliocostalis lumborum muscle predicted holding time in the Sorensen test, while the hip extensor muscles did not contribute. In the present study, in accordance with Mannion and Dolan (1994), two of the most fatigable muscles (multifidi and semitendinosus) were the primary predictors of holding time in the Sorensen test. The result was more ambiguous for the Ito test with one of the least fatigable (iliocostalis) and one of the most fatigable muscles (multifidi) determining holding time. Apart from the fact that the number of study participants was small for the conduction of a multiple regression analysis in the present study as well as in the study by Coorevits et al. (2008), which might impede a conclusive an-

swer to the question which muscles best predicted endurance time, methodological differences might partly explain the controversial results. In the study by Coorevits et al. (2008), the participants performed the modified Sorensen test that had some minor variations (fewer than three straps, arm position at side of the head rather than crossed in front of the chest) to the original protocol (Moreau et al., 2001). Furthermore, the activation of different muscles was measured and electrode placement was not identical. The semitendinosus muscle, which influenced holding time in the present study, was not assessed by Coorevits et al. (2008). Instead, the thoracic parts of some back muscles were measured, which were disregarded in the present study and should be considered in a consecutive study. Nonetheless, in the present study, the facts that in the test position of the Sorensen test, the hip extensor muscles were more activated than in the Ito test and that one hip extensor muscle contributed to holding time in the Sorensen test, but not in the Ito test, might result from differences between the two test positions in terms of position of the pelvis and the lumbar spine and might argue for better criterion validity of the Ito test than of the Sorensen test.

The positions of both tests loaded the muscles below 50% MVC, which is in line with other studies (Moreau et al., 2001; Plamondon et al., 2002). However, the present study involved exclusively young participants. For slim and strong subjects, a muscle activation of 20–25% MVC was proposed in the Sorensen test (Jorgensen and Nicolaisen, 1987), which was exceeded in the present study. Overall, the Sorensen test was more strenuous for the participants than the Ito test. This was reflected in the NMF_{slope} values, which were (numerically) lower throughout the Sorensen test. Test position might thereby be decisive. In the Sorensen test, the arms were in front of the chest, shifting the center of mass towards the head, resulting in a longer lever arm. Furthermore, the fact that muscle activation of the gluteal muscles, although contracting with more conscious awareness in the Ito test, was not significantly different from the Sorensen test, might indicate differences in the position of the pelvis and the lumbar spine in terms of lordosis between the two tests. A significantly longer lever arm for the back extensor muscles in lordosis resulting in less force needed to produce a certain torque was shown by Tveit et al. (1994). The same might be the case for the hip extensor muscles. Since these muscles are biomechanically and anatomically linked to the thoracolumbar fascia, they can commit to the lumbar extension moment (Champagne et al., 2008).

The participants in the present study were young and healthy, which led to holding times that were in the range of previous studies (Nicolaisen and Jorgensen, 1985; Jorgensen and Nicolaisen, 1987) or above (Ito et al., 1996; Latimer et al., 1999; Coorevits et al., 2008) and consequently to a high percentage of subjects who completed all the tests. In patients, however, holding time is significantly decreased (Demoulin et al., 2006). Therefore, the fact that the Ito test seems to load the muscles to a lower degree than the Sorensen test should not be relevant.

In general, the NMF_{slope} values found in the present study were comparable to previous findings (Coorevits et al., 2008) with the values of the multifidi muscles being (numerically) lower (most negative) than any other values. However, it is debatable whether surface EMG is suitable for measuring multifidus muscle activity. While some authors argue in favour of surface EMG (Arokoski et al., 2002), others require intra-muscular recording techniques (Stokes et al., 2003). In addition, the occurrence of crosstalk describing the detection of an EMG signal over a muscle that is generated by another muscle nearby (Schlumberger and Schmidtbleicher, 2000; Freiwald et al., 2007), might have influenced the results of the present study, since multiple muscles close to each other were assessed. Further limitations of the present study were that although test order did not significantly influence test results, there was a tendency towards shorter holding times in the Sorensen test, when the Ito test was performed first. Thus, in a consecutive study, the break between the two tests might be elongated, the activities of the subjects during break should be standardized (preferably in the supine position) and recovery of the fatigued muscles might be assessed either subjectively or objectively. Furthermore, in a consecutive study, one should consider including either men or women. Similarly to other studies (Biering-Sorensen, 1984; Mannion et al., 1997; Kankaanpaa et al., 1998), holding time of women in the present study was longer for both the Sorensen and the Ito test. Due to anthropometric differences (Biering-Sorensen, 1984) and differences in the type I/II muscle fiber ratio of the erector spinae muscle (Mannion et al., 1997), gender differences are relevant in back muscle endurance tests.

In future studies, the Ito test should be further investigated in larger healthy populations as well as in patient groups. Furthermore, the standardization as it was suggested in the present study should be tested for its reliability. Lastly, as proposed in the original study by Ito et al. (1996), lumbar lordosis during the Ito test should be investigated by a radiographic study. Although the Ito test procedure was hypothesized to load the spine less than the Sorensen test (Demoulin et al., 2006), one participant in the present study, who performed the Sorensen test first, terminated the Ito test due to lumbar pain, but did not experience pain during the Sorensen test.

5. Conclusion

The results of the present study showed good criterion validity of the Ito test and indicate that the Ito test might assess back muscle endurance more specifically than the Sorensen test. Thus, the Ito test might present a valuable alternative for testing back muscle endurance in LBP patients. However, these findings have to be verified in consecutive studies with a patient's collective.

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Roland Müller is a movement scientist with an additional background as teacher in physical education who obtained his PhD in Science (2006) from ETH Zurich, Switzerland. He has been working at the Laboratory for Biomechanics of ETH Zurich and at the Research Lab of the Spinal Cord Injury Center of Balgrist University Hospital Zurich, Switzerland. Since 2007 he is back at ETH Zurich as a Senior Researcher in the Institute of Human Movement Sciences and Sport where he leads the Movement Analysis and Rehabilitation Group. The main focus of his work in research and teaching is related to movement analysis and motor control with a special emphasis on work with healthy and handicapped children.

Kurt Strässle received his master degree in human movement sciences from the ETH Zurich, Switzerland, in 2010. The present study was his master thesis.

Brigitte Wirth is a physiotherapist (1996) who obtained her master degree in human movement sciences from the ETH Zurich, Switzerland, in 2004 and her PhD in neuroscience from the ETH Zurich in 2008. She is now working as a Senior Researcher in the Unit Motor Control and Learning of the Institute of Human Movement Sciences and Sport at the ETH Zurich, where she is involved and supervises various projects in orthopedic and neurologic rehabilitation. Her main interests are the fields of assessment and intervention strategies.